

PSCA CAR V δ 1 T cells: a safer off-the-shelf CAR T therapy for pancreatic cancer?

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To cite: Fraietta JA, Lohmueller JJ. PSCA CAR V δ 1 T cells: a safer off-the-shelf CAR T therapy for pancreatic cancer? *Journal for ImmunoTherapy of Cancer* 2026;**14**:e014088. doi:10.1136/jitc-2025-014088

Accepted 20 April 2026



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ABSTRACT

Pancreatic cancer remains among the deadliest malignancies, with a 5-year survival of ~13%. Chimeric antigen receptor (CAR) T-cell therapy offers hope, but conventional $\alpha\beta$ T cells can trigger severe toxicities, including graft-versus-host disease (GvHD) when used for allogeneic therapy. By contrast, $\gamma\delta$ T cells recognize tumors without MHC-major histocompatibility complex restriction and are unlikely to cause GvHD, making them attractive candidates for “off-the-shelf” immunotherapy. Li *et al* engineered the V δ 1 subset of $\gamma\delta$ T cells with a CAR targeting prostate stem cell antigen (PSCA) and compared these cells to CAR-engineered V δ 2 $\gamma\delta$ and conventional $\alpha\beta$ T cells in preclinical pancreatic cancer models. All three groups showed comparable tumor killing efficacy, but the CAR V δ 1 T cells induced none of the GvHD or systemic toxicity seen with CAR $\alpha\beta$ T cells. They also displayed lower exhaustion than CAR V δ 2 cells, suggesting potential for greater persistence. V δ 1 CAR T cells could be expanded robustly ex vivo and remained highly potent even after cryopreservation, a key “off-the-shelf” requirement. These findings position CAR V δ 1 T cells as a safer alternative to traditional CAR T cells. Future rigorous clinical evaluation will be needed to confirm superior safety and efficacy of this promising approach in patients.

Pancreatic cancer (PC) is notorious for its poor prognosis, with a mere 13% 5-year survival rate.¹ Unlike leukemias and lymphomas, where chimeric antigen receptor (CAR) T therapies have achieved remarkable success, solid tumors like PC remain difficult targets.² One major hurdle is toxicity. Conventional CAR T cells derived from $\alpha\beta$ T lymphocytes can cause life-threatening complications, including graft-versus-host disease (GvHD) (if donor-derived) and cytokine release syndrome (CRS).³ These risks, along with the immunosuppressive tumor microenvironment, have limited the translational impact of CAR T cells in cancers such as PC.⁴

$\gamma\delta$ T cells have emerged as a compelling alternative for cellular immunotherapy.⁵ These unconventional T cells combine features of innate and adaptive immunity, recognizing stress-induced ligands on tumors without the

need for antigen presentation by major histocompatibility complex (MHC) molecules (reviewed in⁶). Critically, $\gamma\delta$ T cells do not target allogeneic human leukocyte antigens, so they are far less prone to mediating GvHD even when coming from unmatched donors. This inherent safety advantage positions $\gamma\delta$ T cells as ideal vehicles for allogeneic “off-the-shelf” therapies.⁵ Among $\gamma\delta$ T cells, the V δ 1-expressing subset predominantly resides in tissues (including epithelial tumors), whereas V δ 2 T cells circulate in peripheral blood. The tissue tropism of V δ 1 $\gamma\delta$ T cells suggests they might be especially effective in homing to solid tumor environments. However, V δ 1 cells are rare in blood and historically difficult to expand, so their therapeutic potential has been underexplored.⁶ Until now, most $\gamma\delta$ T-cell trials have used the more abundant V δ 2 T cells, with modest results.⁵

Li *et al*⁷ tackled the challenge head-on by developing a CAR-targeted at prostate stem cell antigen (PSCA), a cell-surface antigen frequently overexpressed in pancreatic adenocarcinoma. They engineered this CAR into human V δ 1 $\gamma\delta$ T cells and achieved robust ex vivo expansion of the otherwise scarce V δ 1 population. In a head-to-head comparison, the authors evaluated PSCA CAR-engineered V δ 1 cells alongside CAR V δ 2 $\gamma\delta$ T cells and conventional CAR $\alpha\beta$ T cells, in functional assays. Notably, all three CAR T-cell types demonstrated comparable antitumor activity against PSCA⁺ PC cells. The CAR V δ 1 T cells were just as potent in killing tumor targets as the familiar CAR $\alpha\beta$ T cells, indicating no sacrifice of efficacy when using $\gamma\delta$ T cells. This finding is noteworthy because it shows that the less common V δ 1 subset can mount tumor cytotoxicity on par with both mainstream $\alpha\beta$ T cells and the V δ 2 subset, at least in controlled models.

Where the V δ 1 CAR T cells truly distinguished themselves was in safety.⁷ In immunodeficient mouse models engrafted with

human tumors, CAR $\alpha\beta$ T cells readily caused GvHD, as expected from their alloreactivity. In contrast, mice treated with PSCA CAR V δ 1 T cells showed no signs of GvHD or cytokine storm, with clinical scores indistinguishable from untreated controls and far better than the CAR $\alpha\beta$ T-cell group. The V δ 1 T cells thus lived up to the immunological promise of $\gamma\delta$ cells by attacking the tumor without attacking the host. It is also notable that the CAR-modified V δ 1 cells did not trigger the systemic inflammatory responses that underlie severe CRS. This may be partly due to the different cytokine profiles of $\gamma\delta$ T cells, but it aligns with early clinical observations that $\gamma\delta$ T-cell therapies tend to be better tolerated than $\alpha\beta$ T-cell therapies. The implication is powerful because an allogeneic CAR T-cell product could be given safely to any patient, without the need for patient-specific manufacturing or heavy immunosuppression to prevent GvHD.

Another important finding by Li *et al.*⁷ relates to T-cell exhaustion. Repeated antigen exposure often drives CAR T cells into a dysfunctional, “exhausted” state, especially in solid tumor settings. Using single-cell RNA sequencing, the authors found that PSCA CAR V δ 1 T cells had significantly lower expression of exhaustion markers compared with CAR V δ 2 T cells.⁷ This result echoes prior reports that tissue-resident V δ 1 T cells can retain function even when expressing exhaustion markers.^{8,9} For therapy, this could mean CAR V δ 1 T cells persist and continue fighting longer than other $\gamma\delta$ T-cell types. A durable effector population is especially crucial in PC, where tumors are adept at suppressing immune cells. While the exact mechanisms remain to be clarified, V δ 1 cells may have intrinsic transcriptional programs that confer greater stamina. If this lower exhaustion translates to better *in vivo* persistence, it would further enhance the therapeutic potential of CAR V δ 1 cells over current alternatives.

From a translational perspective, a key advance of this study⁷ is the demonstration that CAR V δ 1 T cells can be produced at scale and cryopreserved without losing functionality. Both fresh and thawed CAR V δ 1 cells exhibited comparable cytotoxicity *in vitro* and *in vivo*. This means a batch of V δ 1 CAR T cells could be manufactured from a healthy donor, banked, and shipped frozen to treat multiple patients on demand, the paradigm of an “off-the-shelf” cell therapy (figure 1). Achieving this was non-trivial, given the historical difficulty in expanding V δ 1 T cells. Li *et al.*⁷ optimized culture conditions (including feeder cells and cytokines) to selectively propagate V δ 1 T cells from peripheral blood, reaching the scale needed for therapy. The resulting cell product was >95% pure V δ 1 T cells and genetically modified with the CAR, plus a safety switch (truncated epidermal growth factor receptor (EGFR)) for potential clinical use. The ability to bank these cells adds a practical translational edge to the approach because it moves CAR T therapy a step closer to an allogeneic model similar to how we transfuse donor blood or give bone marrow transplants, rather than a bespoke autologous treatment. This is especially relevant

for aggressive cancers like PC, where patients may not have weeks to wait for a personalized cell product. Operationally, this is still an approximately 3–4 weeks process: donor peripheral blood mononuclear cells (PBMCs) undergo 14–21 days of $\gamma\delta$ enrichment, V δ 1 cells are then flow-sorted, re-expanded with weekly irradiated K562 feeder cells, activated for 48 hours, subjected to retroviral vector-mediated CAR transfer, further expanded, and finally cryopreserved for later use.⁸ That feeder-supported, retroviral workflow is important practical context for future Good Manufacturing Practices (GMP) translation and cost modeling.

While Li *et al.*⁷ showcase a compelling proof-of-concept, the preclinical context leaves key translational questions unanswered. All efficacy and safety data were obtained in NSG mice lacking a fully immunocompetent tumor microenvironment, so it remains unclear how PSCA CAR V δ 1 $\gamma\delta$ T cells will perform under the immunosuppressive pressures and antigen heterogeneity of human pancreatic tumors. For example, CAR V δ 1 cells exhibited significantly lower exhaustion marker expression than their V δ 2 counterparts by transcriptomic analysis, but the authors acknowledge that this putative durability advantage must be confirmed in more physiologic models. Similarly, PSCA is not uniformly expressed in PC (ie, present in 60–80% of tumors), and it is uncertain whether the innate tumor-recognition mechanisms of V δ 1 T cells can compensate for antigen-negative escape, particularly since those non-CAR interactions appeared to play only a minimal role in this model. While receptors such as NKG2D, DNAM-1, and NKp30 were found to be up-regulated, blocking them on CAR V δ 1 cells in co-culture assays did not affect cytotoxicity. These gaps are all addressable, and they point to logical next steps such as evaluating CAR V δ 1 cells in immunocompetent or patient-derived models and elucidating V δ 1 T-cell receptor ligand interactions, to ensure this promising therapy can overcome tumor microenvironmental barriers and antigen escape in patients. A useful intermediate step would be donor-matched PBMC-humanized dual-MHC-knockout NSG mice, which attenuate xenogeneic GvHD while preserving human T-cell function and extend the window for testing residual alloreactivity and inflammatory toxicity.¹⁰

In summary, Li *et al.*⁷ have provided evidence that V δ 1 $\gamma\delta$ T cells can be engineered into potent CAR T cells for solid tumor therapy without the usual safety liabilities. Their PSCA-targeted CAR V δ 1 cells matched the antitumor efficacy of conventional CAR T cells while avoiding GvHD and CRS, and showed signs of greater durability via reduced exhaustion. Equally important, the study established methods to expand and preserve these cells, supporting the feasibility of an “off-the-shelf” allogeneic product. Taken together, the work positions PSCA CAR V δ 1 T cells as a promising new modality for PC immunotherapy, potentially an equally efficacious but safer alternative to CAR $\alpha\beta$

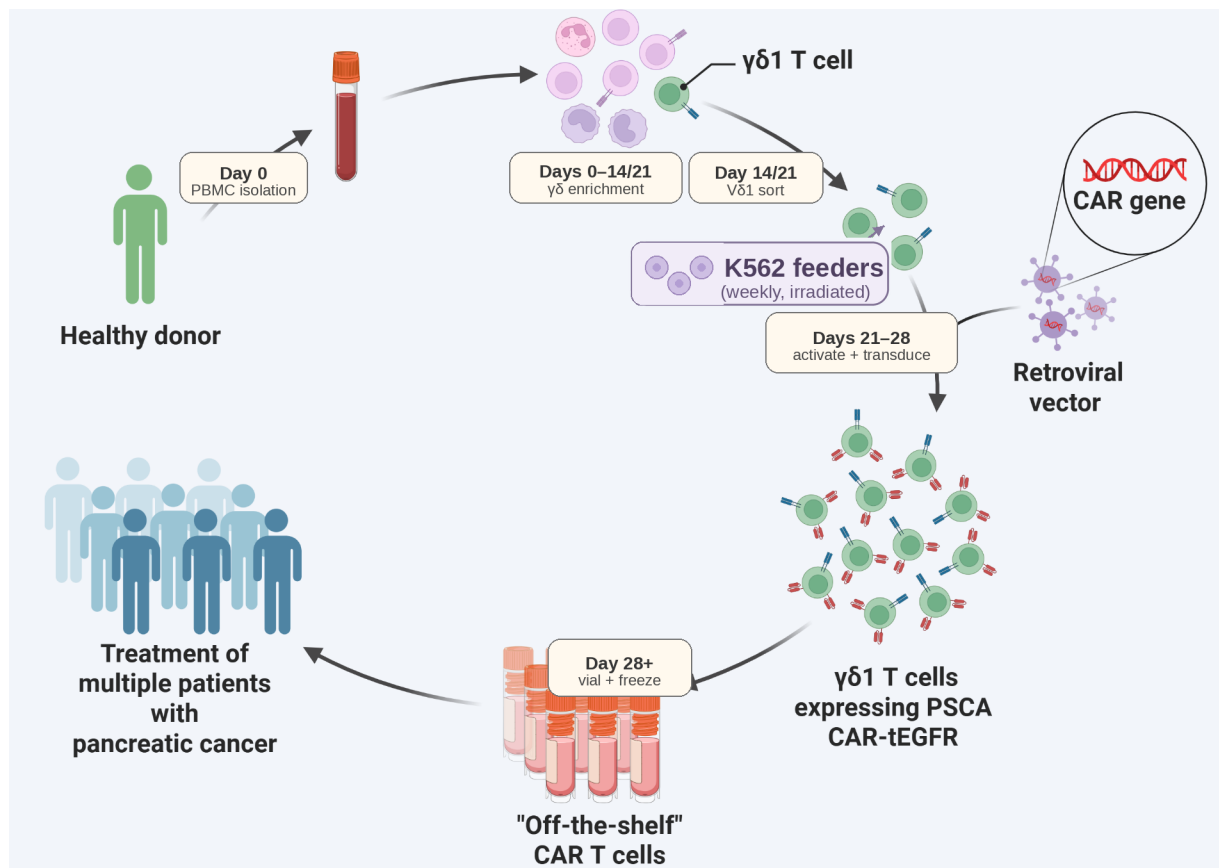


Figure 1 Workflow for “off-the-shelf” PSCA CAR-V δ 1 $\gamma\delta$ T cells. Schematic summarizes the allogeneic manufacturing concept evaluated by Li *et al.* Peripheral blood mononuclear cells (PBMCs) from a healthy donor are enriched for V δ 1 $\gamma\delta$ T cells, expanded, and transduced with a retroviral vector encoding a PSCA-specific chimeric antigen receptor (CAR) linked to a truncated EGFR (tEGFR) safety marker. The CAR⁺ V δ 1 $\gamma\delta$ T cells are further expanded, vialled, and cryopreserved to generate an off-the-shelf product that can be thawed for treatment of multiple recipients with PSCA⁺ pancreatic cancer. Figure generated with BioRender.com. Approximate day annotations are included to indicate the overall engineering timeline, with $\gamma\delta$ enrichment over 14–21 days followed by V δ 1 sorting, feeder-supported expansion, activation, retroviral transduction, and cryopreservation. Weekly irradiated K562 feeder-cell support is shown explicitly in the revised schematic. CAR, chimeric antigen receptor; EGFR, epidermal growth factor receptor; PBMC, peripheral blood mononuclear cell; PSCA, prostate stem cell antigen; tEGFR, truncated EGFR.

T cells. This strategy exemplifies the broader push in the field toward universal cell therapies that can be delivered readily and safely to patients. Of course, the true test will be in clinical trials. Early clinical results with V δ 1 $\gamma\delta$ CAR T cells targeting CD20 for relapsed/refractory (R/R) B-cell non-Hodgkin’s lymphoma have shown limited toxicities with an absence of GvHD, as well as promising persistence and efficacy. Questions remain about how these cells will perform in the immunosuppressive human solid tumor microenvironment and whether their advantages will translate into meaningful patient outcomes. Nonetheless, this study is an encouraging step forward. It highlights how harnessing the unique biology of V δ 1 $\gamma\delta$ T cells could overcome some of the long-standing barriers in CAR T therapy. The path is now set for early-phase trials to evaluate whether PSCA CAR V δ 1 T cells can indeed fulfill the promise as a safe, “off-the-shelf” treatment against one of the hardest-to-treat cancers.

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Contributors JF and JJJ conceived the study, drafted and finalized the manuscript, and managed its submission. All authors approved the final version. JF and JJJ serve as the guarantors of the work.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

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